

CS & IT ENGINEERING



Operating System

CPU Scheduling

Lecture - 01

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Recap of Previous Lecture



Topic

Process states

Topic

Process State Transition

Topic

Process Scheduling

Topic

Types of Schedulers

Topics to be Covered



Topic

Process Scheduling

Topic

Process Scheduling Algorithms

Topic

FCFS Scheduling

Topic

SJF Scheduling

#Q. Consider a system with n processes and m CPUs. Maximum and Minimum number of processes in each of the following states possible? $(n > m)$

1. Ready state $\Rightarrow n$
2. Running State $\Rightarrow m$
3. Blocked State $\Rightarrow n$

⇒ Long-term scheduler Controls the degree of multiprogramming
(L.T.S.)

L.T.S. maintains a good mix of I/O bound & CPU
bound processes

⇒ As degree of multiprogg. increases, CPU utilization also increases.
but upto certain limit.



Topic : CPU Scheduling



Function:

- Make a selection *of process to run next on CPU*

Goal

- Minimize Wait time and Turn-around time
- Maximize CPU utilization (Throughput)
- Fairness



Topic : CPU Scheduling Types

Non-preemptive

Preemptive



Topic : Scheduling Times

{ Assumption $\Rightarrow$ all processes do not have any I/O requirement }

- Arrival Time (AT): The time at which the process is admitted.
- Burst/Service Time (BT): The amount of time a process runs on CPU
- Completion Time (CT): The time at which process completes execution
- Turn-Around Time (TAT): The amount of time a process spends from arrival to completion. $TAT = CT - AT$
- Waiting Time (WT):

Total time spent by process in ready queue for waiting for CPU.

$$WT = TAT - BT$$



Topic : Scheduling Times

- Response Time (RT): Time taken to get first response of process after arrival
- Scheduling Length (L): from when to when scheduling is done.
$$L = \max(CT) - \min(AT)$$
- Throughput:
no. of process executed per unit time
$$= \frac{n}{L}$$

 $n = \text{no. of processes executed}$



Topic : Scheduling Algorithms



1. ✓ FCFS
2. ✓ SJF
3. ✓ SRTF
4. HRRN ✗
5. ✓ Priority Based
6. ✓ Round Robin
7. Multilevel Queue Scheduling
8. Multilevel Feedback Queue Scheduling



Topic : FCFS (First Come First Serve)

Scheduling Criteria: smallest AT process first

Tie breaker $\Rightarrow$ smallest process id first

Type of Algorithm: non-preemptive



Topic : FCFS (First Come First Serve)

Process	Arrival Time	Burst Time
P1	0	24
P2	0	3
P3	0	3

Gantt chart:- (starts from 0 always)





Topic : FCFS (First Come First Serve)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time	R.T.
P1	0	24	24	24	0	0
P2	0	3	27	27	24	24
P3	0	3	30	30	27	27

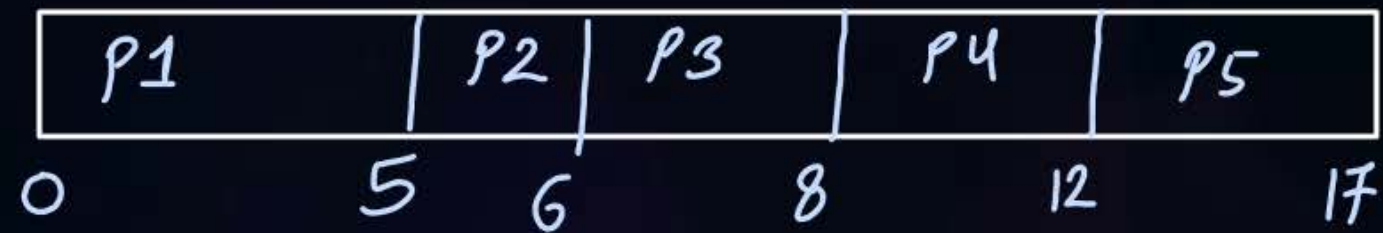
$$\text{avg TAT} = \frac{24 + 27 + 30}{3} = 27$$

$$\text{avg WT} = \frac{0 + 24 + 27}{3} = 17$$



Topic : FCFS (First Come First Serve)

Process	Arrival Time	Burst Time	CT	TAT	WT
P1	0	5	5	5	0
P2	1	1	6	5	4
P3	2	2	8	6	4
P4	3	4	12	9	5
P5	4	5	17	13	8



$$\text{avg TAT} = \frac{38}{5} = 7.6$$

$$\text{avg WT} = \frac{21}{5} = 4.2$$



Topic : FCFS (First Come First Serve)

Process	Arrival Time	Burst Time	CT	TAT	WT
P1	5	4	9	4	0
P2	8	2	21	13	11
P3	6	3	12	6	3
P4	3	1	4	1	0
P5	1	2	3	2	0
P6	7	7	19	12	5



$$\% \text{ of time CPU idle} = \frac{2}{21} * 100\%$$

$$\% \text{ of time CPU utilized} = \frac{19}{21} * 100\%$$



2 mins Summary

Topic

Process Scheduling

Topic

Process Scheduling Algorithms

Topic

FCFS Scheduling

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SJF Scheduling



Happy Learning

THANK - YOU